

Experiment 5

Develop a program that executes the equivalent of the shell command: `ls -l | grep ".c"` using `fork()`, `pipe()`, `dup2()`, and `exec()` system calls.

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int fd[2];

    // Create pipe
    if (pipe(fd) == -1)
    {
        perror("pipe");
        exit(1);
    }

    // Create first child process
    pid_t pid1 = fork();

    if (pid1 == -1)
    {
        perror("fork");
        exit(1);
    }

    if (pid1 == 0)
    {
        // Child 1: execute ls -l

        close(fd[0]);          // Close read end

        dup2(fd[1], STDOUT_FILENO); // stdout -> pipe

        close(fd[1]);
```

```
    execlp("ls", "ls", "-l", (char *)NULL);

    perror("execlp ls");
    exit(1);
}

// Create second child process
pid_t pid2 = fork();

if (pid2 == -1)
{
    perror("fork");
    exit(1);
}

if (pid2 == 0)
{
    // Child 2: execute grep ".c"

    close(fd[1]);          // Close write end

    dup2(fd[0], STDIN_FILENO); // pipe -> stdin

    close(fd[0]);

    execlp("grep", "grep", ".c", (char *)NULL);

    perror("execlp grep");
    exit(1);
}

// Parent closes both pipe ends
close(fd[0]);
close(fd[1]);

// Wait for both child processes
waitpid(pid1, NULL, 0);
waitpid(pid2, NULL, 0);
```

```
return 0;
```

```
}
```

```
yaswanth6970@Yash6970:~$ nano exp5_b.c
yaswanth6970@Yash6970:~$ gcc exp5_b.c -o exp5_b
yaswanth6970@Yash6970:~$ ./exp5_b
-rw-r--r-- 1 yaswanth6970 yaswanth6970 93 Jul 31 09:33 beforefork.c
-rw-r--r-- 1 yaswanth6970 yaswanth6970 833 Aug 31 04:57 exp1.c
-rw-r--r-- 1 yaswanth6970 yaswanth6970 1051 Aug 31 05:01 exp2.c
-rw-r--r-- 1 yaswanth6970 yaswanth6970 874 Aug 31 05:08 exp3.c
-rw-r--r-- 1 yaswanth6970 yaswanth6970 260 Jul 31 10:00 exp4.c
-rw-r--r-- 1 yaswanth6970 yaswanth6970 783 Aug 31 05:10 exp4_wait.c
-rw-r--r-- 1 yaswanth6970 yaswanth6970 867 Aug 31 05:11 exp4_waitpid.c
-rw-r--r-- 1 yaswanth6970 yaswanth6970 1235 Aug 31 05:15 exp5_b.c
-rwxr-xr-x 1 yaswanth6970 yaswanth6970 16128 Aug 18 03:12 filecall
-rw-r--r-- 1 yaswanth6970 yaswanth6970 283 Aug 18 03:21 filecall.c
-rw-r--r-- 1 yaswanth6970 yaswanth6970 80 Jul 28 04:09 helloworld.c
-rw-r--r-- 1 yaswanth6970 yaswanth6970 493 Aug 18 03:47 read.c
-rw-r--r-- 1 yaswanth6970 yaswanth6970 28 Aug 31 05:03 source.txt
-rw-r--r-- 1 yaswanth6970 yaswanth6970 559 Aug 31 05:12 zombie.c
-rw-r--r-- 1 yaswanth6970 yaswanth6970 692 Aug 31 05:13 zombie_fixed.c
```